Klinik Medinara

Operational Performance & Efficiency - Klinik Medinara

Transforming Operational Healthcare Data into Strategic Business Decisions

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Tools Used :



Klinik Medinara: Business Context

Company

Multi-specialty clinic featuring 25 doctors across 8 specialties :

- General Practice
- Pediatrics
- Internal Medicine
- Dentistry
- Dermatology
- OB-GYN
- ENT
- Ophthalmology

Dataset

- appointments
- patients
- doctors
- diagnoses
- prescriptions

Data Period: June 2021 – January 2025 (≈3.5 years).

Context

Curb revenue leakage, optimize physician productivity, and enhance service efficiency.

5 Key Business Questions to Answer

FINANCIAL

How much revenue is lost to cancellations and no-shows?

OPERATIONAL

What is the payment mix composition, and are there insurance mismatches?

VALUES

Which doctors and specialties are the most productive?

RETENTION

What is the ratio of returning vs. new patients, and where are they located?

PHARMACY

What are the most frequent diagnoses and drug prescription patterns?

Approach & Methodology



Dataset

Data Preparation & Validation

Exploratory Data Analysis (EDA)

Dashboard Modeling & Calculation

Business Insights & Recommendation

Insight 1 : Quantifying Lost Revenue

Realized Revenue

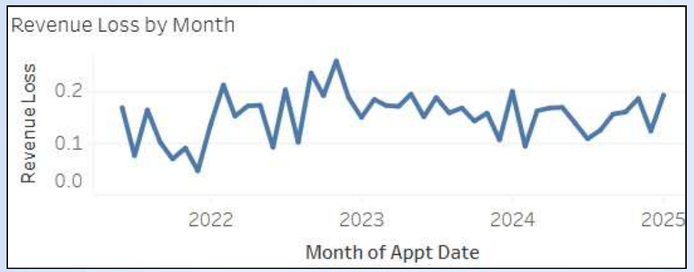
Rp736,44M

Revenue Loss IDR

Rp135,87M

% Revenue Loss

14,76%



Key Findings

Almost 15% of our total potential revenue is simply **lost** because **patients cancel or fail to show up.**
Dr. Iwan Anggara (Dermatology) generates the **2nd highest total revenue**, yet has the **highest no-show rate (10.8%)**.

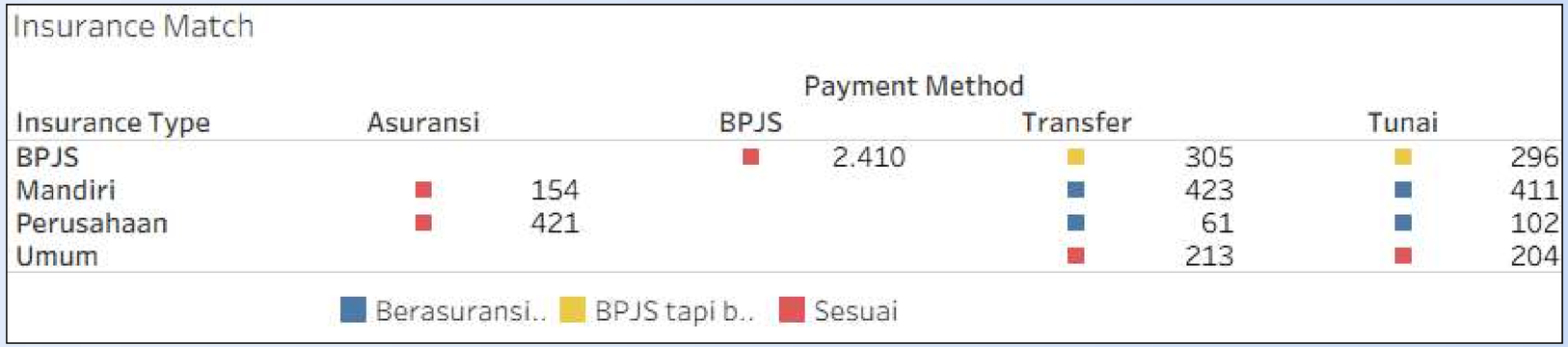
Business Impact

Equivalent to **losing** approximately **2 full months** of **revenue** from a top-performing doctor each year.

Recommendation

Implement automated **H-1 WhatsApp appointment reminders** targeting these top 3 high-no-show specialties. Estimated potential recovery: Rp31M/year if no-shows are cut in half.

Insight 2 : Payment Composition & Insurance Discrepancies



Key Findings

- **12%** of appointments for registered **BPJS patients** were settled via Out-of-Pocket **Cash/Transfer** instead of BPJS claims.
- **84%** of patients with **private insurance** paid out-of-pocket **rather** than filing an **insurance claim**.

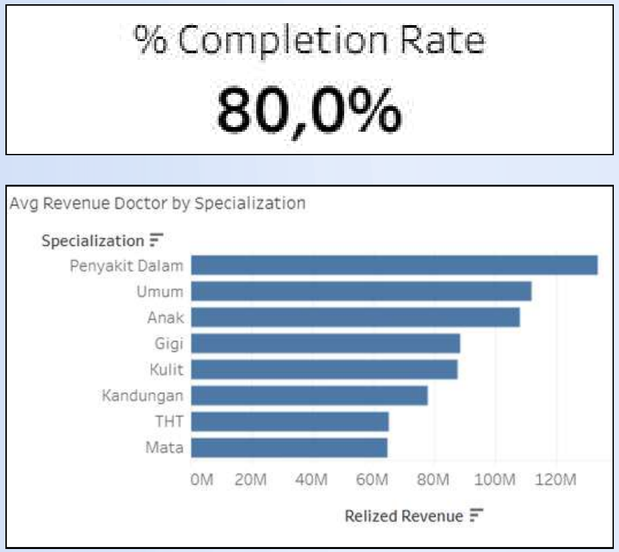
Business Impact

Signals inefficiencies in front-desk eligibility verification during check-in, potentially **delaying claim-based cash inflows.**

Recommendation

Audit and streamline front-desk check-in SOPs for **BPJS/private insurance verification.**

Insight 3 : Identifying Top-Performing Doctors & Specialties



Key Findings

- **Internal Medicine** achieves the **highest revenue** yield per physician despite a **low doctor count.**
- **Differences** in **appointment** volume among doctors are relatively **small (±200 appointments/doctor).**

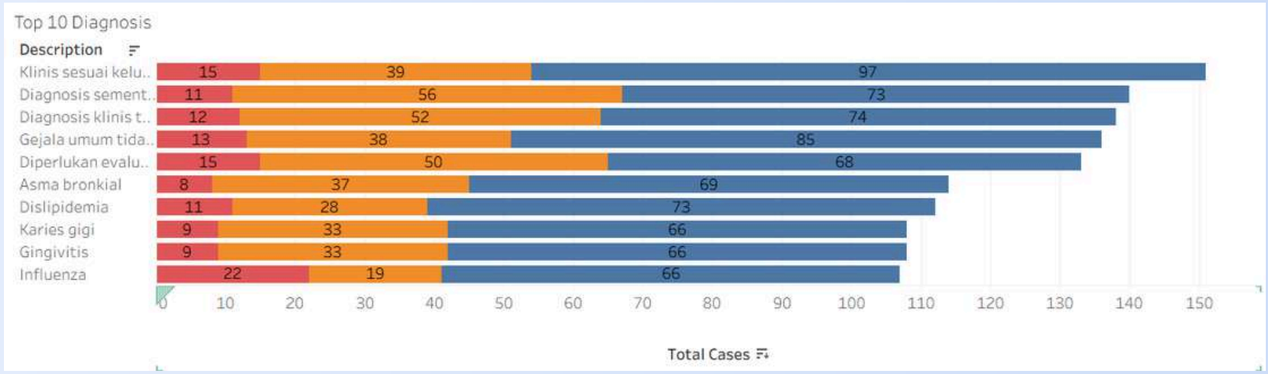
Business Impact

Signals inefficiencies in front-desk eligibility verification during check-in, potentially **delaying claim-based cash inflows.**

Recommendation

Re-evaluate booking slots and **schedule** allocations for doctors falling **below the completion average.**

Insight 4 : Primary Diagnoses & Medication Patterns



Key Findings

Diagnosis distribution is evenly spread across categories with no single dominant disease.

Business Impact

Informs procurement **planning** to maintain balanced **pharmacy stock** and **specialty-level staffing.**

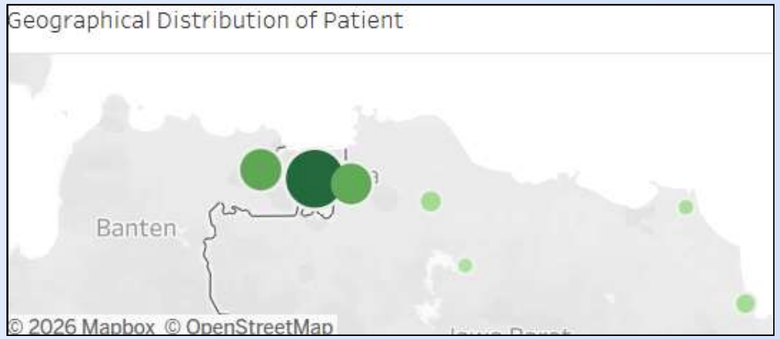
Business Impact

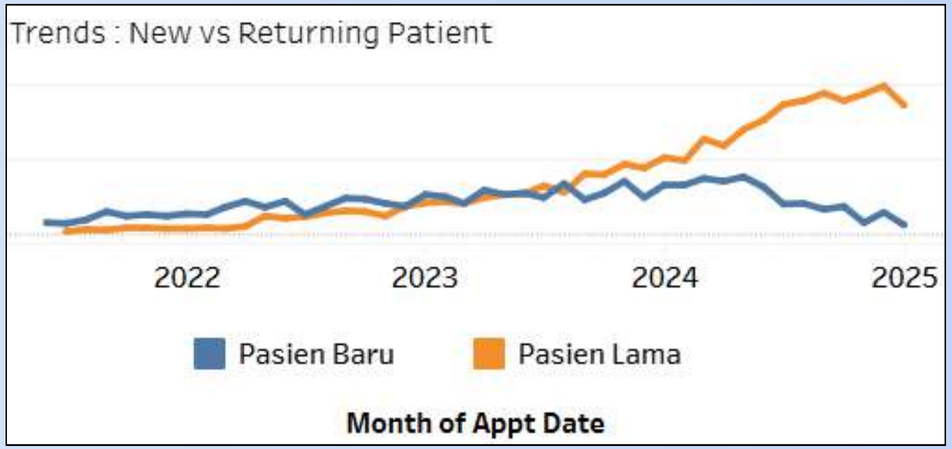
Align monthly pharmacy procurement proportionally with disease severity levels (**60% mild, 30% moderate, 10% severe cases).**

Insight 5 : Patient Loyalty & Geographical Distribution

Repeat Patient Rate %

78.86%





Key Findings

The vast majority of appointments come from **returning patients**, indicating a strong loyalty base.

Business Impact

Opportunity for **hyper-targeted marketing** in **minor contributing cities** while strengthening retention programs in **core hubs.**

Recommendation

Launch formal patient **loyalty programs** to preserve repeat rates, and conduct feasibility studies for expanding service coverage into peripheral suburban areas.

5 Insights into Strategic Action

Key Insight

- 14.8% Revenue Leakage concentrated heavily in Ophthalmology, Dermatology, and ENT.
- Insurance Mismatch: 12% of BPJS and 84% of private insurance patients pay out-of-pocket instead of utilizing insurance claims.
- High Specialty Yield: Internal Medicine generates the highest average revenue per doctor.
- Balanced Disease Load: Diagnoses are evenly distributed with no single condition dominating volume.
- High Retention Base: Strong patient loyalty with a high repeat patient rate.

Recomended Action

- Rp135.9M Loss Exposure → Automated H-1 WhatsApp reminders for high-no-show specialties (est. Rp31M/year saved).
- Cashflow Discrepancies → Mandatory check-in verification audits for BPJS and private insurance.
- Doctor Capacity Variance → Adjust schedule booking slots and workload distribution by specialty.
- Spread Out Pharmacy Demand → Implement proportional stock procurement tailored to specialty diagnosis trends.
- Strong Loyal Patient Base → Introduce retention rewards and explore targeted marketing expansions in satellite cities.

“Transforming Operational Healthcare Data into Strategic Business Decisions.”

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Dashboard Klinik Medinara | Tableau Public